

Shrunken Reality: Augmenting Real-World Contexts in Real-Time on Realistic Miniature Dioramas

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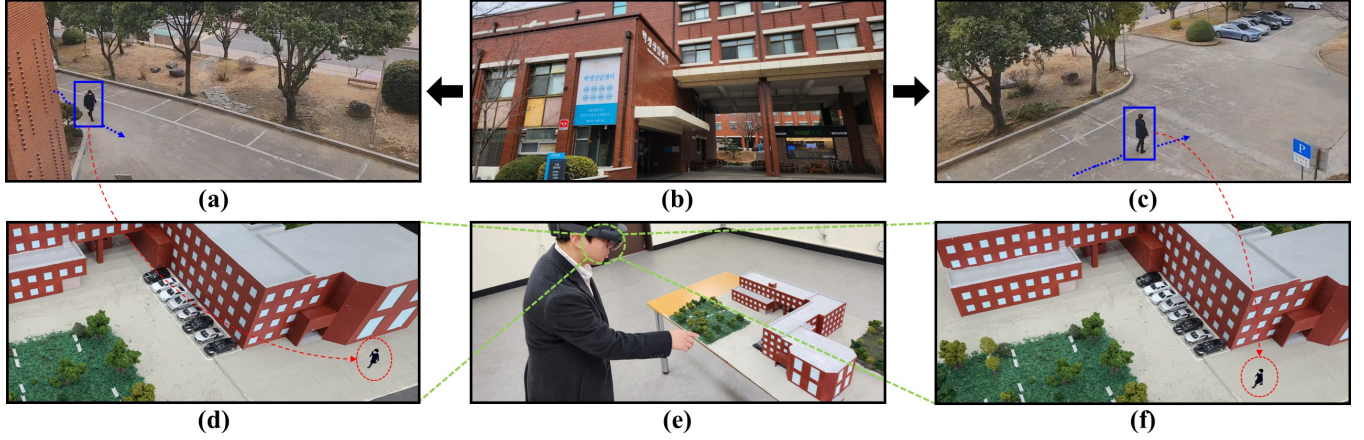


Figure 1: Overview of shrunken reality: real-world pedestrians are augmented onto a miniature diorama viewed by the user. (a) CCTV image at t_1 . (b) Real world. (c) CCTV image at t_2 . (d) User view at t_1 . (e) Shrunken world with user. (f) User view at t_2 .

Abstract

This study presents a new form of augmented reality (AR), referred to as shrunken reality (SR), that captures various real-world contexts in real-time using cameras and augments them onto realistic miniature dioramas, denoted as shrunken worlds (SWs), providing a distinctive experience different from existing AR technologies. We explored the mathematical and technical methods to realize SR, focusing on the processes involved in mapping and augmenting real-world contexts onto miniature, scaled-down models. By exploring the integration of physical miniaturization with digital augmentation, we investigated the potential applications of SR across various sectors, highlighting its relevance to both public services and industries. Finally, we discuss future research directions, such as increasing the number of cameras to capture comprehensive real-world contexts, utilizing deep learning models to enhance user's immersive experience, improving the naturalness of the augmented overlays on SWs, and broadening the scope of augmentations to cover a wider range of real-world, real-time contexts.

CCS Concepts

• Computing methodologies → Mixed / augmented reality.

Keywords

augmented reality, mixed reality, miniature dioramas

ACM Reference Format:

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1 Introduction

Augmented reality (AR) seamlessly integrates virtual content onto a screen, creating an illusion of virtual content existing within the real world (RW). Over the years, the foundational AR technologies, such as computer vision and graphics, have evolved significantly, leading to the development of diverse AR applications for tourism, shopping, education, and various other industries [Lee et al. 2023; Xu et al. 2022; Yuen et al. 2011]. AR integrates virtual content into the real world (RW), offering various benefits, such as enhanced user experience, additional context, real-time updates, interactivity, and simulations. However, AR's expansion is limited because virtual content must be augmented within the user's real-world space or objects. Therefore, we propose a concept called shrunken reality (SR), which expands the definition of AR by augmenting various real-time contexts from the RW onto realistically replicated



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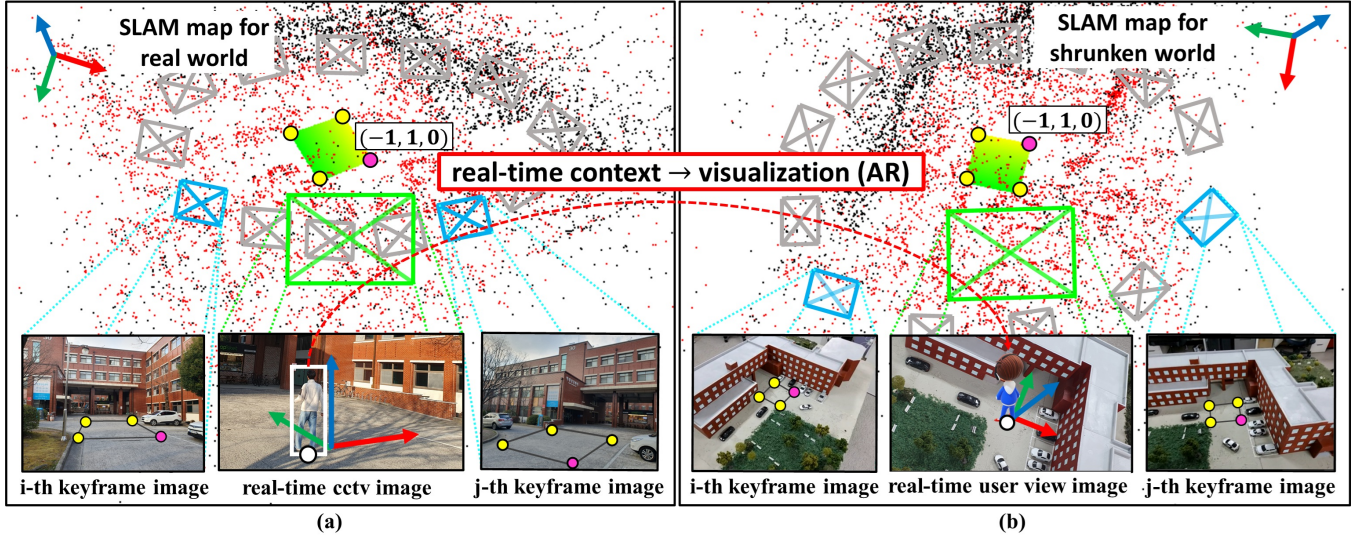


Figure 2: (a) Coordinate systems for the RW: the RW itself, RW ; a SLAM map for the RW, M_{RW} ; real-time CCTV images, I_{CCTV} ; and two keyframe images in M_{RW} , $K^i_{M_{RW}}$ and $K^j_{M_{RW}}$. (b) Coordinate systems for the SW: the SW itself, SW ; a SLAM map for the SW, M_{SW} ; real-time user view images, I_{USER} ; and two keyframe images in M_{SW} , $K^i_{M_{SW}}$ and $K^j_{M_{SW}}$.

miniature dioramas called shrunken worlds (SWs). That is, readily observable contexts in the RW, such as movements of humans and cars, as shown in Figs. 1(a) and (c), can be observed in the SW, as shown in Figs. 1(d) and (f) at times t_1 and t_2 . For example, it can be applied to surveillance systems of apartment complexes, wherein security guards regularly monitor closed-circuit television (CCTV) feeds. However, as shown in Fig. 1(e), simply observing a single SW can yield similar results without the need for extensive surveillance or CCTV monitoring.

2 Methodologies

To implement SR, it is imperative to construct a SW. The straightforward method involves the utilization of architectural blueprints of real buildings. By extrapolating the height dimensions from these blueprints, basic 3D models can be created. Furthermore, to realistically render virtual objects within the SR, these 3D models must be combined to simulate occlusion, ensuring that objects obscured by buildings are appropriately masked out as hidden; this will be discussed in Section 4. The 3D models are then utilized to fabricate scaled-down miniature building using a 3D printer. Fig. 1(d) shows the 3D-printed miniature building derived from the 3D model at a scale of 1:100. Using the physically outputted buildings as a reference, the surrounding real-time contexts are restored to construct the SW. For the SR, contexts are collected from the CCTV cameras installed in the RW, and its geometrical information is transformed into that from the perspective of the user viewing the SW. To execute this process, it is necessary to understand the geometric relationship between the RW and CCTV cameras. Similarly, understanding the geometric relationship between the SW and the user's perspective is essential. This study introduces a simultaneous localization and mapping (SLAM) method [Mur-Artal and Tardós 2017] for this purpose. Additionally, we employ a real-time object

detection algorithm, called you only look once (YOLO) [Redmon et al. 2016], for pedestrian detection to illustrate the concept of SR. To facilitate the understanding of the concept, the geometrical information of pedestrians is simplified to their foot positions.

2.1 Camera Calibrations

Various cameras are introduced for implementing the SR, including the CCTV cameras in the RW, cameras for pre-SLAM mapping process, and the user's AR camera for relocalization. Each camera should be calibrated using a calibration process to determine the intrinsic parameters. Various algorithms are available for camera calibration, and we employed Zhang's method [Zhang 2000].

2.2 Multiple Coordinate Systems

As shown in Fig. 2, the SR involves multiple coordinate systems. The coordinate systems for the RW, the SW, CCTV camera observing the RW, and user camera observing the SW are denoted as RW , SW , $CCTV$, and $USER$, respectively. Additionally, to implement SR, two pre-SLAM mapping processes are required: one for the RW and the other for the SW. Each SLAM map has its own coordinate system, denoted as M_{RW} and M_{SW} , respectively. After the mapping process, it is imperative to save the map files for future relocalization. Typically, the coordinate systems for M_{RW} and M_{SW} are arbitrarily determined during the pre-SLAM mapping process. In contrast, the coordinate systems for RW and SW can be defined according to individual preferences.

2.3 Bridging RW and SW

As mentioned in Section 2.2, the SLAM system establishes its own coordinate system during the mapping process. After loading the stored maps for relocalization, the system calculates the camera pose for each frame based on the coordinate system it has created.

To define the purposes of the *RW* and *SW* and establish the relationships between M_{RW} and *RW*, as well as between M_{SW} and *SW*, a minimum of two keyframe images from the SLAM system's collection are required. These images must capture the known 3D points of *RW* and *SW*, where an individual may arbitrarily determine according to their preference 3D points such as $(-1, 1, 0)$, $(1, 1, 0)$, $(1, -1, 0)$, and $(-1, -1, 0)$. If each of the 3D points is assigned semantically identical positions in both *RW* and *SW*, this enables the transformation between the *RW* and *SW* coordinate systems using an identity matrix I . As indicated by the yellow and pink points in Figs. 2(a) and (b), four pairs of 2D points are selected in the i -th and j -th keyframe images from M_{RW} and M_{SW} , respectively. Each point in the images is a projected point of the 3D points previously determined by the individual. Subsequently, the camera poses within the SLAM systems for each keyframe, $KF_{M_{RW}}^i$, $KF_{M_{RW}}^j$, $KF_{M_{SW}}^i$ and $KF_{M_{SW}}^j$, are acquired. By leveraging the image coordinates and camera poses, triangulation [Hartley and Zisserman 2003] is employed to calculate the corresponding 3D coordinates of these image coordinates with reference to the SLAM systems.

2.4 Bridging Worlds and SLAM Maps

The transformation between worlds and SLAM maps is computed using a 3D-3D correspondence between previously calculated 3D points and the 3D points determined by the individual. The process of determining an optimal similarity transformation matrix between two sets of 3D point clouds, A and B , involves the following steps:

$$t_A = -\frac{1}{N_A} \sum_{i=0}^{N_A-1} p_A^{(i)}, t_B = -\frac{1}{N_B} \sum_{i=0}^{N_B-1} p_B^{(i)} \quad (1)$$

where N_A and N_B are the number of points within A and B , respectively. $p_A^{(i)}$ and $p_B^{(i)}$ denote the 3D coordinates of the i -th point in point clouds A and B , respectively. Initially, the means of these point clouds are computed to translate them into the origin such that only rotation can be considered. The two negative signs indicate that each point cloud moves from its center to the origin.

$$s = \sqrt{\frac{\sum_{i=0}^{N-1} (p_B^{(i)})^T (p_A^{(i)})}{\sum_{i=0}^{N-1} (p_A^{(i)})^T (p_A^{(i)})}}; R_{scaled} = sR \quad (2)$$

A rotation matrix R between point clouds A and B is derived using singular value decomposition (SVD) on the point correspondences. The resulting matrices from the SVD are used to compute R , which is adjusted to prevent reflection when $\det(R)$ is negative. Finally, the scale factor s is computed for the final scaled rotation matrix R_{scaled} .

$$T = \begin{bmatrix} 1 & -t_B \\ 0 & 1 \end{bmatrix} \begin{bmatrix} R_{scaled} & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & t_A \\ 0 & 1 \end{bmatrix} \quad (3)$$

$$[R|t]_{A \rightarrow B} = \frac{1}{T_{33}} T[:3, :4] \quad (4)$$

Matrix T maintains a similar shape between the two point clouds while performing transformations such as translations from A to the origin and from the origin to B , rotation, and scaling. Using the matrix T , the transformation matrix $[R|t]_{A \rightarrow B}$ obtained from *RW* to M_{RW} is denoted as $[R|t]_{RW \rightarrow M_{RW}}$, whereas that obtained from

SW to M_{SW} is denoted as $[R|t]_{SW \rightarrow M_{SW}}$ and are obtained using Eq. (4). Although specific fixed keyframes, such as the i -th and j -th keyframes, as shown in Figs. 2(a) and (b), were used to compute the transformation matrices, these matrices ($[R|t]_{RW \rightarrow M_{RW}}$ and $[R|t]_{SW \rightarrow M_{SW}}$) are permanently applicable across all real-time frames because they no longer rely on cameras such as *CCTV* or *USER*. Additionally, I_{CCTV} and I_{USER} are real-time images of each camera, as depicted with green rectangles in Figs. 2(a) and (b).

2.5 Bridging Worlds and Cameras

The transformation matrix between the *RW* and the *CCTV* looking at the *RW* and that between the *SW* and the *USER* looking at the *SW* are denoted as $[R|t]_{RW \rightarrow CCTV}$ and $[R|t]_{SW \rightarrow USER}$, respectively, which can be defined using the SLAM system.

$$[R|t]_{RW \rightarrow CCTV} = [R|t]_{M_{RW} \rightarrow CCTV} \times [R|t]_{RW \rightarrow M_{RW}} \quad (5)$$

$$[R|t]_{SW \rightarrow USER} = [R|t]_{M_{SW} \rightarrow USER} \times [R|t]_{SW \rightarrow M_{SW}} \quad (6)$$

$[R|t]_{M_{RW} \rightarrow CCTV}$ and $[R|t]_{M_{SW} \rightarrow USER}$ are matrices of extrinsic parameters relative to their respective SLAM coordinate systems and calculated by localizing camera poses in the SLAM process. Note that the *CCTV* cameras are fixed in position; therefore, once the $[R|t]_{M_{RW} \rightarrow CCTV}$ are determined, they remain constant, and accordingly, the $[R|t]_{RW \rightarrow CCTV}$ are also constant. In contrast, the views of the user can change in real-time, so the $[R|t]_{M_{SW} \rightarrow USER}$ is dynamic, resulting in $[R|t]_{SW \rightarrow USER}$ being recalculated for each frame. Moreover, as mentioned in Section 2.4, since $[R|t]_{RW \rightarrow M_{RW}}$ and $[R|t]_{SW \rightarrow M_{SW}}$ apply to all frames, the matrix multiplication of the camera poses calculated by the SLAM system for each frame can be considered applicable for all frames.

3 Results

We verified the validity of the method presented in Section 2.5 by selecting specific 3D coordinates on the *RW* or *SW* and projecting them on a 2D screen. For the sake of experimentation, this study conducted experiments on the *SW*. Let us project the four coordinates with reference to the *RW* defined in Section 2.3, which is the simplest sample. $s[px, py, 1]^T$ is the homogeneous form of image coordinates for the user's screen and s is the scale element resulting from the homogeneous calculation. $[X, Y, Z, 1]^T$ is the homogeneous form of the 3D coordinates with reference to the *SW*.

$$s[px, py, 1]^T_{I_{USER}} = K_{3 \times 3} \times E_{3 \times 4} \times [X, Y, Z, 1]^T_{SW} \quad (7)$$

It is assumed that the internal parameter K of the AR device responsible for the camera projections is already known, based on the prior camera calibration described in Section 2.1. The external parameter E represents the top part (3×4) of $[R|t]_{SW \rightarrow USER}$. As mentioned in Section 2.3, because the transformation matrix from *RW* to *SW* is just an identity matrix I obtained for semantic purposes, the four *RW* points defined in Section 2.3 can be considered to be the same as the *SW* points. Using Eq. (7), we can project the 3D coordinates from *SW* into AR device's 2D screen, as indicated by the blue dots in Fig. 3(b). The ultimate goal of SR is to convert the 2D coordinates of objects, such as pedestrians captured by *CCTV* cameras, into the coordinate system *RW*, and subsequently into the coordinate system *SW*. In other words, conversion from 2D to

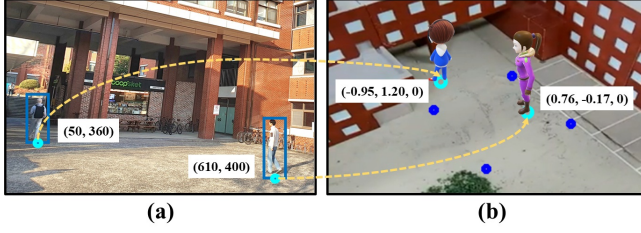


Figure 3: (a) Image coordinates of the context obtained from CCTV in the RW and (b) corresponding 3D coordinates of the context in the SW.

3D is essential using Eq. (8). Additionally, w is a scale element that changes the value of the last element to 1 such that X and Y can be obtained.

$$\text{let } P_{3 \times 4} = K_{3 \times 3} \times E_{3 \times 4} \text{ and } Z = 0. \text{ Then,} \quad (8)$$

$$w \begin{bmatrix} X \\ Y \\ 1 \end{bmatrix}_{RW} = \begin{bmatrix} P_{00} & P_{01} & P_{03} \\ P_{10} & P_{11} & P_{13} \\ P_{20} & P_{21} & P_{23} \end{bmatrix}^{-1} \begin{bmatrix} px \\ py \\ 1 \end{bmatrix}_{I_{CCTV}}$$

Here, the image coordinates of the bottom center of the bounding box returned by YOLO, which detects pedestrians, are utilized as the 2D coordinates px and py . These coordinates must be transformed into 3D coordinates from the perspective of RW . Assuming that the Z coordinates of a pedestrians' feet are zero, X and Y can be calculated. For example, let us consider that YOLO returns the 2D coordinates of a pedestrians' feet, as indicated by the cyan dots in Fig. 3(a). First, the 2D coordinates are transformed into 3D coordinates from the perspective of RW using Eq. (8). Thereafter, they are converted from the RW to SW using the identity matrix I . Subsequently, the 3D coordinates of the pedestrians from the perspective of SW are calculated, as shown in Fig. 3(b). Finally, using the matrix $[R|t]_{SW \rightarrow USER}$, the 3D coordinates of the pedestrians in the SW are projected onto the user's AR device display, as indicated by the cyan dots in Fig. 3(b).

4 Discussion

In this study, only one CCTV camera was used to detect pedestrians for augmentation. It is anticipated that employing multiple cameras at various locations can provide a wider range of observations and more realistic augmentation. However, multiple cameras can result in re-identification issues for multiple pedestrians [Bedagkar-Gala and Shah 2014]. The utilisation of SR in specific fields, such as industry and surveillance systems, is a significant issue. To implement a more realistic augmentation of pedestrian movements, it is necessary to consider not only the 3D locations of pedestrians but also their movement directions. As shown in Fig. 4, the issue of "reality blending," the phenomenon in which pedestrians are augmented in a SR as walking beside, below, or behind buildings, appearing obscured from the user's field of view must be addressed. While this study focused only on pedestrian detection, SR can expand its scope by incorporating a variety of object types, such as cars and pets. Furthermore, the integration of deep learning-based anomaly detection techniques [Pang et al. 2021] could enhance the ability of

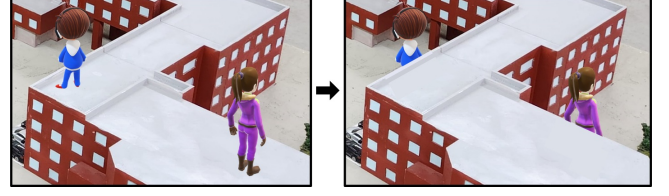


Figure 4: An example of reality blending.

SR to detect abnormal situations in real-world contexts, providing the immersive, realistic AR experience.

5 Conclusion

This study introduced SR, a novel form of AR that leverages the capture of real-world context and miniature physical models to create a unique user experience. Unlike conventional AR, which overlays digital information onto the real world, SR captures real-world contexts through cameras and seamlessly integrates them onto realistic miniature dioramas. By exploring the integration of physical and digital worlds, SR has the potential to enhance public services and benefit various industries. We suggested future research directions, including increasing the number of cameras, using deep learning for accident prevention, improving overlay naturalness in SWs, and broadening the scope of SR to cover broader range of real-time data streams. We believe that SR presented a significant advancement in AR, offering a distinctive and potentially transformative user experience.

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